

House Price Prediction – Project Summary

Overview

The objective of this project was to develop a machine learning model capable of predicting house prices using various property features such as area, number of bedrooms, bathrooms, stories, parking availability, furnishing status, and other amenities. The dataset contained 545 records and 13 attributes, with **price** serving as the target variable.

Data Preparation

The dataset was loaded and explored using Pandas. Data quality checks showed that there were no missing values. Duplicate records were identified and removed where necessary. Categorical features such as furnishing status, air conditioning, basement availability, and preferred area were converted into numerical form using one-hot encoding. The dataset was then split into training and testing sets using an 80:20 ratio.

Model Development and Evaluation

Two regression models were trained and evaluated:

Model	MAE	RMSE	R ² Score
Linear Regression	970,043	1,324,507	0.653
Random Forest Regressor	1,021,546	1,400,566	0.612

Among the two models, **Linear Regression performed better**, achieving an R² score of 0.653. This indicates that the model was able to explain approximately 65.3% of the variation in house prices while maintaining lower prediction errors than the Random Forest model.

Key Findings

Feature importance analysis revealed that the most influential factors affecting house prices were:

- Number of bathrooms
- Air conditioning availability
- Hot water heating
- Preferred area location
- Furnishing status
- Number of stories
- Basement availability

The analysis showed that properties with additional amenities and better locations generally commanded higher prices. The number of bathrooms emerged as the strongest factor influencing house value.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. The Linear Regression model provided the best performance on the dataset. Based on the findings, real estate businesses should emphasize important features such as bathrooms, premium amenities, and preferred locations when marketing properties, as these factors significantly contribute to higher property values.